

# Scaling the time-course of myelosuppression from rats to patients with a semi-physiological model

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**Summary** *Aim* To investigate the potential of a model for chemotherapy-induced myelosuppression to predict the full time-course of myelosuppression in patients based on rat data. *Methods* White blood cell counts were determined in rats after administration of 5-fluorouracil, epirubicin, cyclophosphamide, docetaxel, paclitaxel or etoposide. Pharmacokinetic models were used to predict the concentration-time profile in each rat. A semi-physiological model of myelosuppression was applied to the rat data. The drug-related parameter Slope was allowed to differ between drugs. The analysis was performed in NONMEM VI. Time-courses of myelosuppression in patients were predicted for each drug based on patient pharmacokinetic models, typical system-related parameters previously determined in patients and the rat Slope estimates in the present study. *Results* The semi-physiological model of myelosuppression fit the rat data well and the estimated maturation time in rats (53 h) was approximately half of the previous estimate in patients. The relative difference in Slope estimates for rats and patients based on total drug concentrations ranged between 28% to 8-fold for the six drugs. The differences reduced to 8–37% for all drugs when correcting the rat Slope estimates for species difference in protein binding and in CFU-GM assay sensitivity. *Conclusions* This method for interspecies scaling was successful in predicting the time-course of myelosup-

pression in patients based on rat data. Predictions improved when species differences in protein binding and CFU-GM assay sensitivity were accounted for. The approach appears promising for predicting myelosuppression in patients early in development.

**Keywords** White blood cells · Myelosuppression · Rats · Anticancer drugs · Interspecies scaling · Population pharmacokinetic-pharmacodynamic modeling · Pharmacometrics

## Introduction

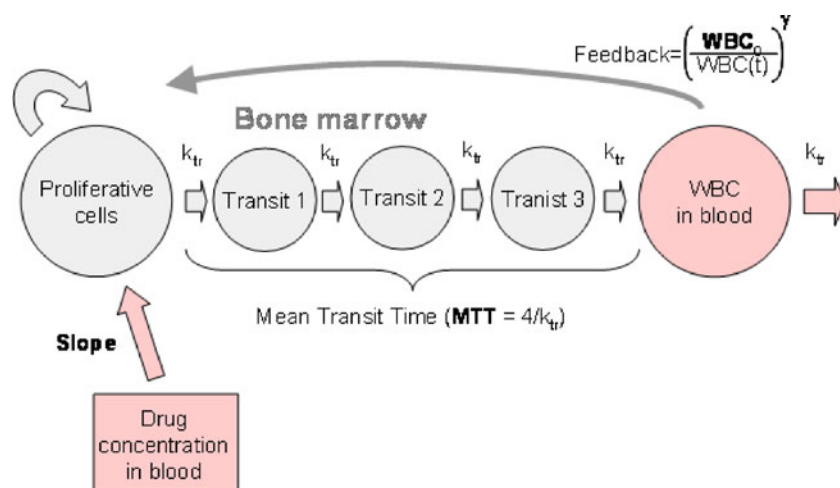
The majority of the classical anticancer drugs have myelosuppression as dose-limiting toxicity and the risk for neutropenia is therefore the major factor determining the dose size that can be administered to patients. Predictions of the full time-course of myelosuppression in patients based on preclinical data would be valuable to early in the development improve on the understanding of drug characteristics and to help prioritize between drug candidates as well as in the development of new treatment regimens of existing drugs.

A semi-physiological model describing the time-course of myelosuppression following anticancer-drug administration in patients has previously been shown to result in similar system-related parameters across drugs [1–10]. The model consists of one compartment with drug-sensitive cells that correspond to the proliferating cells in the bone marrow, three transit compartments of drug-insensitive cells that represent subsequent cell maturation without proliferation and one compartment of the circulating white blood cells (WBC) where the observations are made (Fig. 1). A feedback function based on the number of circulating WBC

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**Fig. 1** The original semi-physiological model of myelosuppression presented in Friberg et al. [1]. Three transit compartments connect the proliferative cells in the bone marrow with the circulating WBC in blood that are observed. A feedback mechanism based on the WBC at baseline ( $WBC_0$ ) and the WBC at time  $t$  ( $WBC(t)$ ) regulates the

proliferative rate. The mean transit time through the maturation chain (MTT) is  $4/k_{tr}$ . The three estimated system-related drug-related parameters are  $WBC_0$ , MTT and the feedback factor ( $\gamma$ ). One drug-related parameter (Slope) describes a linear concentration-effect relationship

in relation to the number of WBC at baseline ( $WBC_0$ ) regulates the rate of proliferation and mimics the effect of endogenous growth factors such as granulocyte colony stimulating factors, G-CSF. For total number of white blood cells, the estimated system-related parameters  $WBC_0$ , mean transit time (MTT) and the feedback factor  $\gamma$  ranged between  $6.74$  to  $8.10 \cdot 10^9/L$ ,  $90$  to  $135$  h and  $0.121$  to  $0.239$ , respectively, for the six investigated drugs docetaxel, paclitaxel, etoposide, DMDC, irinotecan and vinflunine [1]. The drug-related parameter Slope varied between drugs, as expected, since this parameter reflects the potency. It was shown that using fixed, typical values of the three system-related parameters ( $7.0 \cdot 10^9/L$ ,  $125$  h and  $0.17$ , respectively) resulted in similar time-courses as when system-related parameters were estimated separately for each dataset. The model has thereafter been applied to describe myelosuppression in patients of several anticancer drugs, including epirubicin and cyclophosphamide, see e.g. [1–10].

In the work described above parameters were also determined for neutrophils, which is the dominating leukocyte type in humans (60–70%). MTT and  $\gamma$  were similar for both types of data, however as higher Slope estimates (18 to 77 % higher) were obtained for the neutrophil data, neutrophils appear more sensitive to the investigated drugs than other types of white blood cells. The neutrophil counts at baseline were in these studies estimated to be 66 to 77% of total WBC. In rats lymphocytes are the dominating white blood cell (approximately 85% in Sprague Dawley rats) and the absolute number of WBC is higher. Cell turnover rate is thought of being inversely related to body size [11] and therefore MTT is expected to be lower in rats than in patients. The experimentally determined transit time

through the non-mitotic development stages of neutrophils is 3 days in rats [12] and 6.6 days in humans [13]. Despite these quantitative differences rats have been described to be a good species for studying myelosuppression [12, 14].

There are limited studies on scaling of the time-course of drug effects and adverse events from animals to patients. Model-based pharmacodynamic scaling has for example been performed for recombinant human erythropoietin [15], 5-Ht<sub>1A</sub> receptor mediated responses, i.e. hypothermia and cortisol responses [16], respiratory depressant and antinociceptive effects [17], thromboxane B<sub>2</sub> and prostaglandin E<sub>2</sub> dynamics [18] and muscle relaxation [19]. The low number of successful cases of pharmacodynamic scaling may partly be because often parameters used in the scaling are based on empirical models in contrast to physiology-based or mechanistic models. Pharmacokinetic interspecies scaling is however routinely performed and traditionally achieved by allometric principles where flows such as clearance are proportional to body weight by a factor of  $3/4$  and the distribution volume is directly proportional to weight. As pure allometric scaling sometimes have limited predictive capacity, refinement methods based on species-differences in protein-binding and enzyme activity are used to improve the pharmacokinetic predictions.

We hypothesized that for myelosuppression the relative potency between rats and patients may be similar when species-differences are accounted for in the system-related parameters  $WBC_0$ , MTT (cell-turnover) and (potentially)  $\gamma$ . We investigated if the same physiology-based model structure for the time-course of myelosuppression in patients could be applied on leukocyte data obtained from a number of rat studies and if the Slope-estimates derived

from rat data could be used to predict myelosuppression in patients. As only unbound drug is available to exert the effect, the impact of species differences in protein binding was evaluated. In addition, refinements using sensitivity measures from an in vitro assay to improve the predictions, i.e. IC90 ratios in the colony forming unit granulocyte macrophage (CFU-GM) assay, were evaluated as this assay has shown to reasonably well scale the LD10 in mice to MTD in patients for a number of anticancer drugs [20].

## Methods

### Animals

Male Sprague-Dawley rats [CrI:CD(S-D)BR-rats, Charles River, Uppsala, Sweden] were used in all experiments. They were kept in a room illuminated from 7:00 AM to 7:00 PM and had free access to food and water throughout the study. After 1–2 weeks of acclimatization the rats were randomized into different treatment groups. At the first day of treatment the rats weighed on average ( $\pm$ SD) 296 ( $\pm$ 34) g. The Animal Ethics Committee in Uppsala, Sweden, approved the studies.

### Leukocyte counts

Blood (250  $\mu$ l) for determination of number of leukocytes was drawn from a hind paw or thigh vein and sampled in EDTA-prepared Microtainer tubes (Becton Dickinson, Franklin Lakes, NJ). White blood cell counts were determined using a Coulter Counter MD II series (Coulter Electronics Ltd., Luton, England) at baseline (1–2 days before first drug administration) and typically every other day up to 11–15 days after first drug administration. Thereafter cell counts were determined less frequently up to day 30.

### Drugs

Animal experiments were performed where 5-fluorouracil (Flurablastin<sup>®</sup>, NordMedica), epirubicin (Farmorubicin<sup>®</sup>, Pfizer), cyclophosphamide (Sendoxan<sup>®</sup>, Baxter), docetaxel (Taxotere<sup>®</sup>, Sanofi-Aventis), paclitaxel (Taxol<sup>®</sup>, Bristol-Myers Squibb) and etoposide (Vepesid<sup>®</sup>, Bristol-Myers Squibb) were administered in monotherapy. Rats were dosed intraperitoneally (i.p.) or intravenously in to the femoral vein (i.v.). All studies included control rats that received injections of normal saline and were exposed to the same sampling scheme as those rats receiving active drug. As there was no influence of this treatment on the WBC, the WBC data from control rats were not included in the modeling analysis.

### 5-Fluorouracil

The 5-fluorouracil (5-FU) data has previously been described [21, 22]. In brief, in 53 rats the pharmacokinetics and white blood counts were studied following administration of a single dose of 127 mg/kg ( $n=6$ ), 100 mg/kg ( $n=5$ ), 50 mg/kg ( $n=6$ ) or 33 mg/kg ( $n=5$ ), or, as two doses of 50 mg/kg each with a 6 h interval ( $n=6$ ), as three doses with a 4 h interval, each of 33 mg/kg ( $n=6$ ) or of 17 mg/kg ( $n=6$ ), or, as two doses of 63 mg/kg each with a 2 day interval ( $n=6$ ), or as three doses of 49 mg/kg, each with a 2 day interval ( $n=6$ ). Individual pharmacokinetic parameters of a 1-compartment model with non-linear elimination have previously been determined in these rats and were used in the present analysis to predict individual concentration-time profiles [22]. Blood samples for determination of hematological toxicity were drawn at baseline and up to day 11 [21] or day 25 [22].

### Epirubicin

The majority of the epirubicin data used in the modeling has previously been described [21]. In 34 rats the pharmacokinetics and WBC were studied following administration of a single i.p. dose of 7 mg/kg ( $n=6$ ), 3.5 mg/kg ( $n=6$ ) or 2.3 mg/kg ( $n=5$ ), or as two i.p. doses of 3.5 mg/kg each with a 6 h interval ( $n=6$ ), or as three i.p. doses of 2.3 mg/kg ( $n=6$ ) or 1.7 mg/kg ( $n=5$ ), each with a 4 h interval. In addition to these data, 12 rats were administered epirubicin i.v. at single doses of 7 mg/kg ( $n=3$ ), 3.5 mg/kg ( $n=3$ ), 5 mg/kg ( $n=3$ ) or 2.5 mg/kg ( $n=3$ ). The pharmacokinetics was described by a three-compartment model with a first-order absorption compartment and a 71% bio-availability for rats that were administered epirubicin by the i.p. route [21, 23]. Individual pharmacokinetic parameters were available for the rats that received epirubicin i.p. [21]. In the i.v. rats typical population pharmacokinetic parameters were used [23]. Blood samples for determination of hematological toxicity were drawn at baseline and at predetermined time points up to 11–30 days after the day of epirubicin administration.

### Cyclophosphamide/4-hydroxycyclophosphamide

In 34 rats the pharmacokinetics of the active metabolite 4-hydroxycyclophosphamide (4-OHCP) concentrations and WBC were measured following administration of cyclophosphamide in a single i.p. dose of 120 mg/kg ( $n=4$ ), 90 mg/kg ( $n=6$ ), 55 mg/kg ( $n=6$ ), 20 mg/kg ( $n=6$ ), or as three i.p. doses of 30 mg/kg each with a 4 h interval ( $n=6$ ), or as three i.p. doses of 20 mg/kg with a 2 day interval between each dose ( $n=6$ ). The 4-OHCP pharmacokinetics in these rats have previously been described with a one-compartment model with first-order absorption [24]. Individual parameters

were here used to predict the concentration-time profile. Blood samples for determination of hematological toxicity were drawn at baseline and up to 11–30 days.

#### *Docetaxel*

In 20 rats WBC counts were studied following administration of a single i.v. dose of 8 mg/kg ( $n=3$ ), 7 mg/kg ( $n=8$ ), 4 mg/kg ( $n=5$ ), or 2.5 mg/kg ( $n=4$ ). Typical population pharmacokinetic parameters from a 2-compartment model determined in a previous i.v. study in our lab were used to predict the drug concentration-time profile [23]. Blood samples for determination of hematological toxicity were drawn at baseline and up to 22–28 days after docetaxel administration.

#### *Paclitaxel*

In 6 rats WBC counts were determined following administration of a single i.v. dose of 4 mg/kg. Typical population pharmacokinetic parameters based on unbound concentrations were derived from a 2-compartment model on paclitaxel in a study in similar rats (data on file). Blood samples for determination of hematological toxicity were drawn at baseline and thereafter approximately every other day up to 17 days after drug administration with an additional sample on day 28.

#### *Etoposide*

In 11 rats WBC counts were studied following administration of a single i.v. dose of 10 mg/kg. The concentration-time profile was described by pharmacokinetic parameters derived from a previous study in Sprague Dawley rats, from which a 2-compartment model was derived [25]. Blood samples for determination of hematological toxicity were drawn at baseline and up to 28 days after administration.

#### Data analysis

The WBC data from all drugs were analyzed simultaneously using the first-order conditional estimation (FOCE) method in NONMEM VI [26]. The INTERACTION option was applied when appropriate. The blood count data were log-transformed as this approach resulted in a more symmetrical distribution of residuals around zero compared with untransformed data. The objective function value, OFV, was used to compare the fit of two nested models where a drop in OFV of 3.84 corresponds to  $p<0.05$  for a one parameter difference. The R-based program Xpose 4 (<http://xpose.sourceforge.net/>) was used for graphical visualization of the model fit that also guided the model building. PsN (<http://psn.sourceforge.net/>) was used for

executing simulations and calculations for visual predictive checks.

#### Model building

The semi-physiological model of myelosuppression developed based on patient data [1] was applied to the rat data. System-related parameters were here estimated to be the same for all drugs. Drug-specific linear concentration-effect relationships were utilized. To account for the in rats observed decrease in WBC already at day 1 and 2 after drug administration, different modifications of the drug-effect model were tested. Additive, proportional and combined additive and proportional residual error models were assessed. Between-rat variability was evaluated in  $WBC_0$ , MTT and Slope, as well as in the residual error magnitude. For Slope, a common variability parameter for all drugs was used.

#### Model evaluation

The final model was evaluated by numerical [27] and visual predictive checks. Prediction intervals with 95% confidence intervals were obtained by simulating 500 data sets from the model using the original data set as a template.

#### Scaling

The drug-related parameters (Slope values) estimated here for rats were compared with those previously estimated for the same drugs in patients [1–3]. For 5-FU, no Slope-value has been reported in patients and therefore the Slope-estimate was derived by testing which value that resulted in similar nadir as those reported in two published studies [28, 29] when typical system-related parameter values of  $7 \cdot 10^9$  cells/L, 125 h and 0.17 were used for  $WBC_0$ , MTT and  $\gamma$ , respectively.

The estimated Slope values in rats were subsequently corrected for previously observed species differences in (1) protein binding and (2) bone marrow sensitivity between rodents and humans in the CFU-GM assay ( $IC_{90\text{human}}/IC_{90\text{mice}}$ ) [20]. Thereby it was assumed that rats and mice have more similar bone marrow sensitivity to the anticancer drugs than they have with patients, which has indeed been shown to be the case for primary lymphocytes [30].

For 5-FU, epirubicin and cyclophosphamide (and its metabolite), the unbound fraction ( $f_u$ ) is moderate or high (0.2–0.9) and no major species differences have been demonstrated [31–37]. Therefore the  $f_u$  ratio ( $f_{u\text{rat}}/f_{u\text{human}}$ ) was in the scaling set to 1 for these drugs. For docetaxel a  $f_u$  of 0.135 in rats [38] and 0.0661 in patients [39] were applied. For paclitaxel, when administered with Cremophor EL as in

the Taxol® formulation, the unbound fraction in plasma is dependent on the plasma concentration of Cremophor EL [40, 41]. Unbound concentrations were used in the modeling for paclitaxel but when calculating a Slope-estimate based on bound concentrations typical fu values following Taxol® administration of 0.11 in rats (data on file) and 0.05 in patients [40, 41] were used. For etoposide a 10-fold difference in the unbound fraction between rats and humans have been reported [42]; 0.52 and 0.049, respectively. The fu corrected Slope ( $Slope_{fu}$ ) was calculated from the Slope estimated in rats ( $Slope_{rat}$ );

$$Slope_{fu} = Slope_{rat} \cdot (fu_{human}/fu_{rat})$$

The earlier determined sensitivity ratios ( $IC_{90_{human}}/IC_{90_{mice}}$ ) were 5.98, 0.926, 0.27, 1.19, and 0.912 for 5-FU, doxorubicin, cyclophosphamide, paclitaxel and etoposide, respectively [20]. Epirubicin was in the current analysis assumed to have the same  $IC_{90}$  ratio as its stereoisomer doxorubicin, 4-OHCP was assumed to have the same  $IC_{90}$  ratio as its parent drug cyclophosphamide and docetaxel was assumed to have the same  $IC_{90}$  ratio as the other taxane, paclitaxel. The sensitivity corrected Slope ( $Slope_{sensitivity}$ ) was calculated as

$$Slope_{sensitivity} = Slope_{rat}/(IC_{90_{human}}/IC_{90_{mice}})$$

The sensitivity and fu corrected Slope ( $Slope_{fu,sensitivity}$ ) was computed as

$$Slope_{fu,sensitivity} = Slope_{rat} \cdot (fu_{human}/fu_{rat})/(IC_{90_{human}}/IC_{90_{mice}})$$

Predictions of the time-course of myelosuppression were made for a 5-FU bolus dose of 1500 mg/m<sup>2</sup>, an epirubicin bolus dose of 75 mg/m<sup>2</sup>, a cyclophosphamide bolus dose of 600 mg/m<sup>2</sup>, a docetaxel 1-hour infusion of 100 mg/m<sup>2</sup>, a paclitaxel 3-hour infusion of 175 mg/m<sup>2</sup> and a 3-day etoposide infusion of 375 mg/m<sup>2</sup>. The typical system-related parameters in patients ( $WBC_0=7.0 \cdot 10^9/L$ ,  $MTT=125$  h and  $\gamma=0.17$ ) earlier determined [1] were used in the predictions.

## Results

The original model structure developed from patient data [1] described the rat data from all treatments adequately when analyzed simultaneously (Fig. 2a, left panel) and the parameters were estimated with good precision (Table 1). Despite that the system-related parameters ( $WBC_0$ ,  $MTT$  and  $\gamma$ ), IIV parameters and residual error were forced to be the same for all drugs, the model showed acceptable VPCs of the median time course of myelosuppression for all six investigated drugs (Fig. 2b). The original model structure could however not well capture the rapid decline in white blood cells during the first couple of days after drug

administration that were observed in the rat experiments. The decline could be better described by allowing a fraction of the drug effect on the proliferative cells to affect the cells in the transit compartments and in the blood compartment (Fig. 2, right top panel). In this “additional sensitivity model” the Slope estimates were 27% to 88% higher than in the original model (Table 1), i.e. a higher drug potency was estimated and  $MTT$  was estimated to be longer. The linear drug affect on the cells in the transit compartments and the circulating cells (SFR) was estimated to be 39% of the Slope estimate on the proliferating cells (a common value for all drugs was estimated). The WBC nadir was however not better characterized with the additional sensitivity model (Fig. 2a, right panel) and to have a consistent model structure with the one developed earlier for patients the parameters from the original model was chosen when comparing the parameters across species. IIV was included for  $WBC_0$ ,  $MTT$  and Slope (a common IIV value for all drugs). For the original model a proportional error was sufficient to describe the residual error but for the additional sensitivity model inclusion of an additive residual error improved the fit significantly.

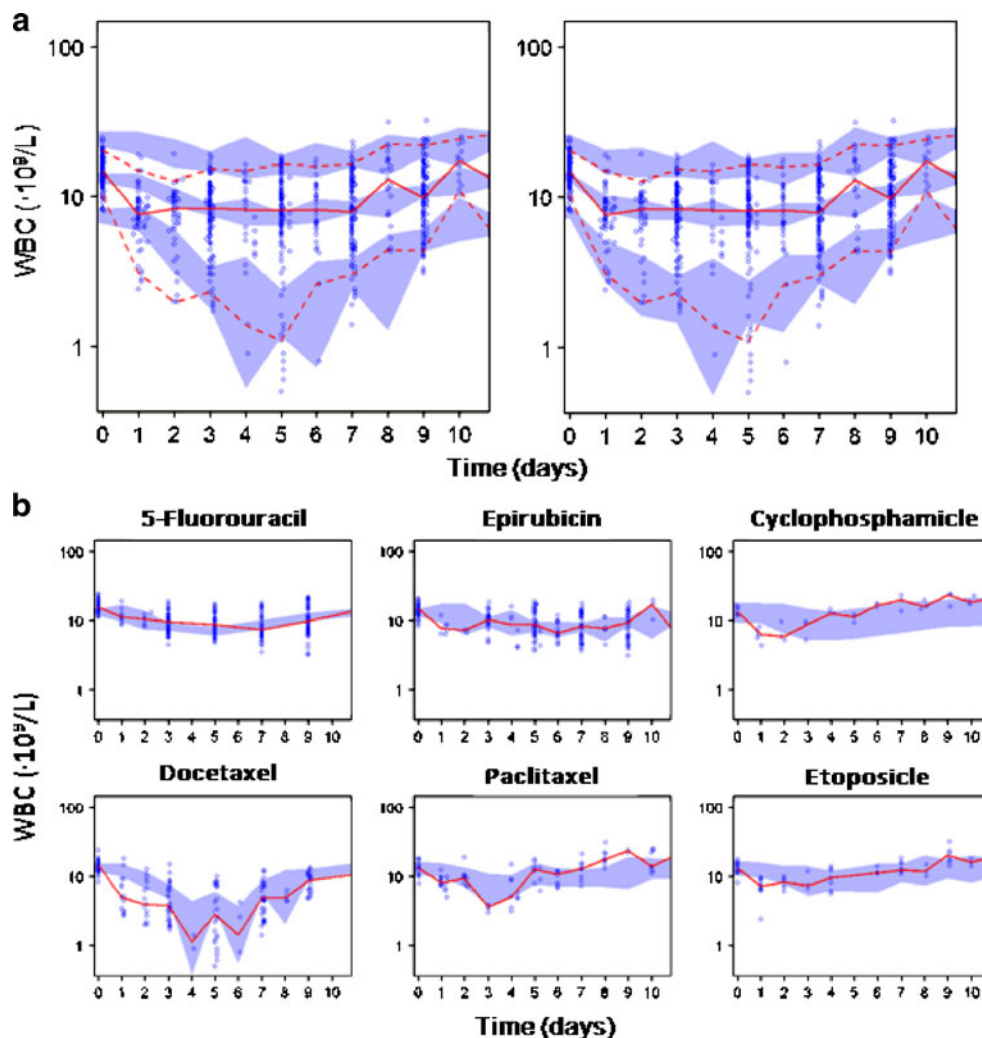
The relative difference in Slope estimates for rats ( $Slope_{rats}$ ) and patients ( $Slope_{patients}$ ) in the original model varied from 28% for epirubicin to a factor of 7–8 for 5-FU, etoposide and 4-OHCP (Table 2, Fig. 3, left panel). For 5-FU and 4-OHCP correction by species differences in  $IC_{90}$  ratio in the CFU-GM assay clearly reduced the species differences in Slope and the corrected values ( $Slope_{sensitivity}$ ) differed to the observed patient Slope values by only 37% and 16%, respectively. For etoposide scaling of Slope values estimated in rats with species differences in protein binding and in vitro sensitivity ( $Slope_{fu,sensitivity}$ ) reduced the relative difference compared to  $Slope_{patients}$  to 8.6%. Following both protein binding and  $IC_{90}$  ratio correction all drugs had relative differences ( $Slope_{fu,sensitivity}$  vs.  $Slope_{patients}$ ) between 8.6% and 37% (Fig. 3, right panel).

Predictions of the typical time course of myelosuppression in patients based on the model, typical pharmacokinetic parameters in patients, typical system-related myelosuppression parameters in patients and predicted Slope values from preclinical data ( $Slope_{fu,sensitivity}$ ) were for both scenarios similar to those previously estimated from patient data (Fig. 4).

## Discussion

A semi-physiological model of myelosuppression was here shown to be applicable to describe the time-course of rat leucocytes following administration of six different drugs. Prediction of clinical grade of neutropenia from in vitro data has previously been made [43, 44], however, this is the

**Fig. 2** Visual predictive checks from 500 simulated data sets for **a** data from all drugs for the original model (left panel) and for the model where all compartments were assumed to be drug sensitive (right panel) and for **b** each drug separately for the original model. Observed leukocytes (o), and the 5th, 50th and 95th percentiles of the observed data (— and ---) with the 95% confidence intervals of the corresponding percentiles (shaded areas) are shown. The limited data for each drug only supports medians to be shown in b



**Table 1** Objective function value (OFV) and parameter estimates (% relative standard error, RSE) for WBC in rats from the original model and the additional sensitivity model where part of the drug effect (SFR) affected all transit compartments and the circulating compartment

|                                        | Original model |                      | Additional sensitivity model |                      |
|----------------------------------------|----------------|----------------------|------------------------------|----------------------|
|                                        | Typical (RSE%) | IIV, CV% (RSE%)      | Typical (RSE%)               | IIV, CV% (RSE%)      |
| OFV                                    | -1912.0        | —                    | -2278.4                      | —                    |
| WBC <sub>0</sub> (·10 <sup>9</sup> /L) | 13.3 (2.5)     | 23 (5.5)             | 13.6 (2.2)                   | 24 (5.0)             |
| MTT (h)                                | 52.8 (9.1)     | 20 (15)              | 72.9 (3.4)                   | 20 (8.5)             |
| γ                                      | 0.149 (14)     | —                    | 0.196 (5.9)                  | —                    |
| Slope <sub>5-FU</sub> (L/mg)           | 0.247 (14)     | 27 (19) <sup>a</sup> | 0.356 (9.0)                  | 26 (13) <sup>a</sup> |
| Slope <sub>Epirubicin</sub> (L/mg)     | 12.6 (10.3)    | 27 (19) <sup>a</sup> | 18.7 (7.6)                   | 26 (13) <sup>a</sup> |
| Slope <sub>4-OHCP</sub> (L/mg)         | 2.65 (6.9)     | 27 (19) <sup>a</sup> | 3.36 (5.4)                   | 26 (13) <sup>a</sup> |
| Slope <sub>Docetaxel</sub> (L/mg)      | 16.1 (13)      | 27 (19) <sup>a</sup> | 22.6 (8.4)                   | 26 (13) <sup>a</sup> |
| Slope <sub>uPaclitaxel</sub> (L/mg)    | 32.9 (25)      | 27 (19) <sup>a</sup> | 61.7 (15)                    | 26 (13) <sup>a</sup> |
| Slope <sub>Etoposide</sub> (L/mg)      | 1.08 (16)      | 27 (19) <sup>a</sup> | 2.01 (14)                    | 26 (13) <sup>a</sup> |
| SFR                                    | —              | —                    | 0.388 (7.0)                  | —                    |
| Additive error (·10 <sup>9</sup> /L)   | —              | —                    | 0.622 (34)                   | —                    |
| Proportional error (%)                 | 28.1 (4.0)     | —                    | 23.0 (6.4)                   | —                    |

<sup>a</sup> a common Slope IIV for all drugs was estimated

**Table 2** Estimated Slope values from rat and patient data using the original model and predicted Slope values in patients based on rat Slope estimates, species differences in fu and/or human/mice sensitivity (IC90 ratio) in the CFU-GM assay [20]

|                                    | Estimated Rat Slope  | Patient Slope predicted from rat and in vitro data |                              |                                 | Estimated Patient Slope   |
|------------------------------------|----------------------|----------------------------------------------------|------------------------------|---------------------------------|---------------------------|
|                                    | Slope <sub>rat</sub> | Slope <sub>fu</sub>                                | Slope <sub>sensitivity</sub> | Slope <sub>fu,sensitivity</sub> | Slope <sub>patients</sub> |
| Slope <sub>5-FU</sub> (L/mg)       | 0.247                | 0.247                                              | 0.0413                       | 0.0413                          | 0.030 <sup>a</sup>        |
| Slope <sub>Epirubicin</sub> (L/mg) | 12.6                 | 12.6                                               | 13.6                         | 13.6                            | 17.5 <sup>b</sup>         |
| Slope <sub>4-OHCP</sub> (L/mg)     | 2.65                 | 2.65                                               | 9.82                         | 9.82                            | 11.6 <sup>c</sup>         |
| Slope <sub>Docetaxel</sub> (L/mg)  | 16.1                 | 7.88                                               | 13.5 <sup>d</sup>            | 6.62 <sup>d</sup>               | 7.91 <sup>e</sup>         |
| Slope <sub>Paclitaxel</sub> (L/mg) | 3.75 <sup>f</sup>    | 1.65                                               | 3.15                         | 1.38                            | 1.69 <sup>e,g</sup>       |
| Slope <sub>Etoposide</sub> (L/mg)  | 1.08                 | 0.102                                              | 1.18                         | 0.112                           | 0.121 <sup>e</sup>        |

<sup>a</sup> estimated from reports on nadir WBC following 5-FU administration [28, 29]<sup>b</sup> from Sandström et al. 2005 [2]<sup>c</sup> from Sandström et al. 2006 [3]<sup>d</sup> using relative sensitivity estimate from paclitaxel<sup>e</sup> from Friberg et al. 2002 [1], converted from  $\mu\text{M}^{-1}$  to L/mg<sup>f</sup> calculated from estimated Slope<sub>u</sub> using an fu of 0.114 (data on file)<sup>g</sup> calculated from estimated Slope<sub>u</sub> using an fu of 0.05 [40]

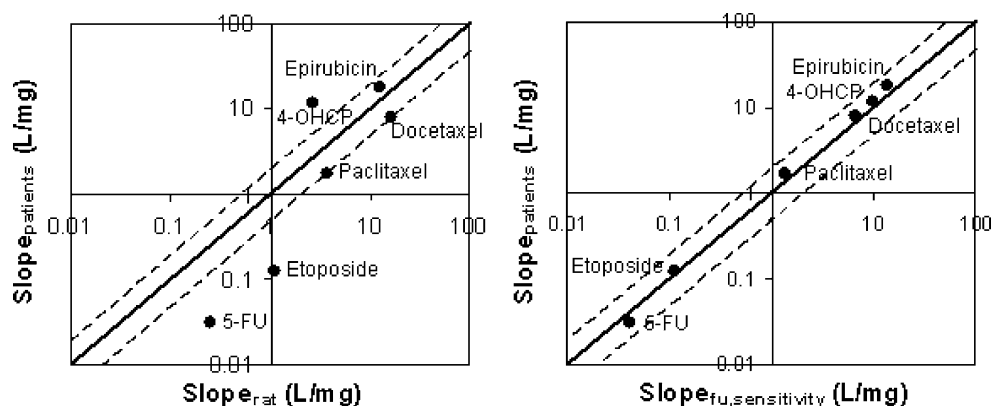
first time the full time-course of myelosuppression has been shown to be predictable from preclinical data. Following correction of previously observed species differences in protein binding and in vitro sensitivity, the relative difference in Slope based on rat (Slope<sub>fu,sensitivity</sub>) and patient (Slope<sub>patients</sub>) data was less than 40% for all drugs and the predicted time-courses of myelosuppression were close to those earlier determined in patients (Fig. 4). For allometric scaling of pharmacokinetics a difference in predicted and observed clearance with a factor of 2 is viewed to be an accurate prediction [45]. The success is likely because the prediction was based on a model built on knowledge of the underlying physiological system and that this system is similar in rats and man. Based on the fact that some maturation and proliferation occurs for circulating lymphocytes, in contrast to neutrophils, it was recently shown that species differences in sensitivity of anticancer drugs for primary lymphocytes are similar to what have been found in the CFU-GM assay [30]. This less labor intensive

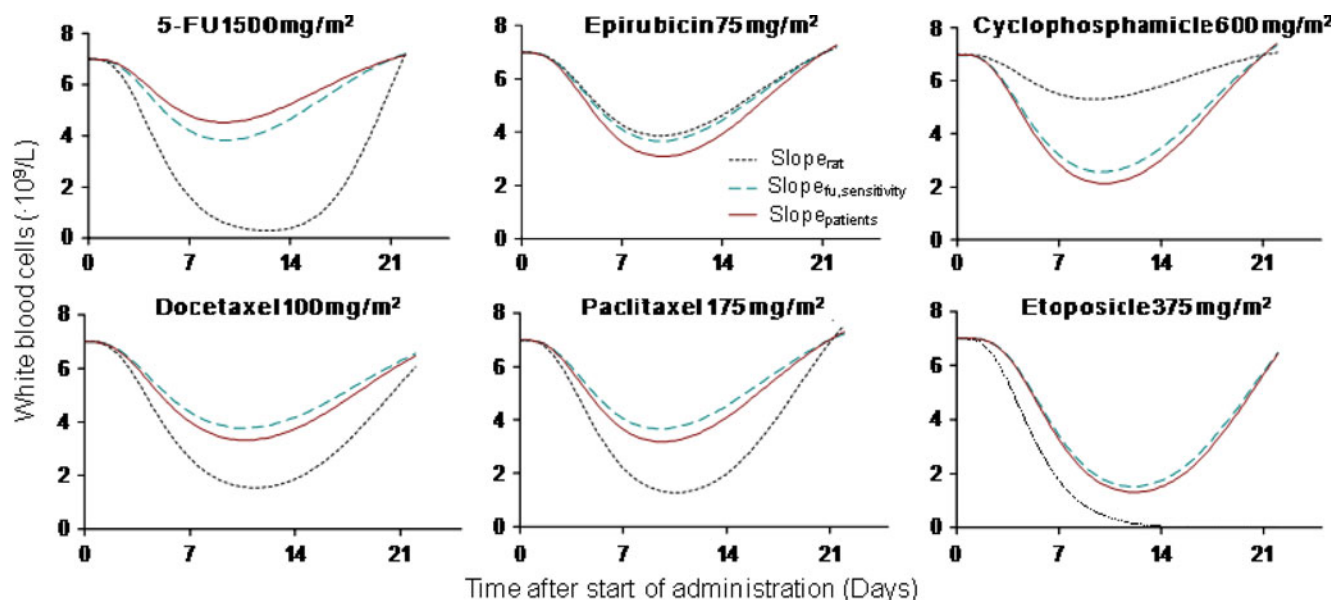
lymphocyte assay may therefore be a good alternative to the CFU-GM assay to evaluate species differences in sensitivity and when scaling Slope<sub>rat</sub> to Slope<sub>sensitivity</sub> values.

In the present study only the effects on WBC were scaled as only WBC data were available in the rat studies. The semi-physiological model of myelosuppression [1] has earlier been applied on both rat neutrophils [46] and patient neutrophils [6, 10] following administration of topotecan. In line with the larger sensitivity of topotecan in patients compared with rats [30], the Slope estimate was lower in rats (39 L/mg) than in patients (133–143 L/mg) in these previous reports.

In the present study the approach for interspecies scaling was to evaluate if the semi-physiological model could be applied on rat data from several drugs and if the drug-related parameters (Slope values) were similar when the species-differences in system-related parameters (WBC<sub>0</sub>, MTT and  $\gamma$ ) had been considered. System-related param-

**Fig. 3** Estimated Slope values in patients (Slope<sub>patients</sub>) based on previous analyses versus estimated Slope values in rats (Slope<sub>rat</sub>; left panel) and predicted Slope values based on the estimates from rat data adjusted for species differences in protein binding and IC90 ratios in the CFU-GM assay (Slope<sub>fu,sensitivity</sub>; right panel). Solid lines are line of identity and dashed lines represents a deviation from the line of identity with a factor of 2





**Fig. 4** Predicted time-courses of myelosuppression in patients when typical system-related parameters in patients were used;  $WBC_0 = 7 \cdot 10^9/L$ ,  $MTT = 125$  h and  $\gamma = 0.17$ , and different Slope values estimated in rats (with and without protein binding and sensitivity scaling), and in patients

eters were however not scaled here as those have previously been characterized in patients and shown to be similar across drugs. Based on allometric principles biological time periods,  $t$ , are dependent on a scaling factor in the power function of  $1/4$  ( $t = \theta \cdot \text{body weight}^{1/4}$ ) [11]. Although there has been a clear correlation to body weight in scaling pharmacodynamic models [15–19] the estimates of the exponent of the correlation have in most cases differed from the theoretical value of  $1/4$ . Interestingly, the exponent for red blood cell and red blood cell precursor life spans were estimated to 0.124–0.148 [15], which is similar to the exponent value of 0.158 that would be used to scale the here estimated MTT for white blood cells of 52.8 h in a 300 g rat to a value of 125 h in a 70 kg patient. The estimated feedback factor  $\gamma$  was 0.149 in rats, a value that is in the same range as those earlier determined in patients (0.121–0.239) [1]. The similarity in  $\gamma$  between the two species is expected as this factor is dimensionless and may imply that the regulation of production following myelosuppression occurs with a similar mechanism in rats and patients.

All available rat WBC count data from all six drugs were estimated simultaneously to estimate typical system-related parameters in rats that could be used as prior information for future data sets, for example when the data is sparse. This approach resulted however in that the fit of the full time-course was not optimized for each of the drugs (Fig. 4). However here the focus was to estimate a typical Slope value for each drug and the median at nadir was well characterized (Fig. 3).

The model fit significantly improved when the drugs were allowed to also affect the transit compartments and the

compartment of circulating cells. This early drug effect on the observed WBC was not observed in the patient data, likely because in patients neutrophils are the dominating white blood cell type and these cells are not proliferating in the last stages of maturation and in the circulation. For lymphocytes, i.e. the dominating white blood cell type in rats, some proliferation is ongoing also among the mature cells that can explain why also cells in the transit (maturing) compartments and the circulating compartments appeared to be sensitive to the studied cytotoxic drugs. The Slope estimates were similar for the two presented models and despite that the additional sensitivity model could account for the early drop in WBC, the Slope-estimates from the original model was chosen for scaling so that the same model structure was used in rats and patients. In both cases the IIV in baseline was overpredicted (Fig. 2), however this is likely because of the assumption that the variability is constant over time, i.e. also after the recovery phase.

In conclusion, scaling of the full time-course of myelosuppression from rat data to patients using a semi-physiological model was successful for six different drugs, particularly after correcting for species differences in protein binding and sensitivity differences determined in the CFU-GM assay. Predictions of myelosuppression are valuable in drug development for the selection of compounds and may also be useful to assess the myelosuppressive effect of treatments of new drug combinations. This approach may also be applicable for evaluation of different drug administration schedules based on rat data. As variability in the parameters between patients has been shown to be similar across drugs [1] simulations of myelosuppression based on rat Slope values and patient

system-related parameters may be performed with adequate accuracy.

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